CSE 413 — Class Performance Test 3 - 7B

Topic: Random Number Generation, Probability Distributions & Visualization
Duration: 30 mins
Mode: Individual, Code-based

Problem: Service Counter Simulation

A university cafeteria wants to study the time students spend waiting at a service counter.

You are asked to simulate the waiting time using Python and NumPy.

Tasks

Part A — Uniform Distribution

Generate **1,000 random waiting times** following a uniform distribution between **3 and 15 minutes**.

1. Store the generated values in a NumPy array.
2. Display the first 10 generated values.
3. Calculate and print:
    - Mean
    - Median
    - Standard deviation

Part B — Normal Distribution

Generate **1,000 waiting times** following a normal distribution with:

- Mean = **9 minutes**
- Standard deviation = **2 minutes**

1. Generate the data using NumPy.
2. Replace any negative waiting time with **0**.
3. Calculate and print:
    - Mean
    - Median
    - Standard deviation
    - Maximum waiting time

Part C — Visualization

Create **one figure containing two subplots**:

- Subplot 1 → Histogram of the uniform-distribution data
- Subplot 2 → Histogram of the normal-distribution data

Each plot must have:

- Appropriate title
- X-axis label
- Y-axis label

Part D — Reproducibility

Run the following experiment:

np.random.seed(32)

Generate a **3×3 matrix of random integers between 1 and 50**.

Then:

1. Run the code again with the same seed.
2. Run it without setting the seed.
3. Print the outputs.
